

ROUSSEL Victor

Paris Area, France

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Second-year engineering student at CentraleSupélec with a strong interest in applied mathematics, particularly in AI, data science, and cryptography. Actively seeking internships in these fields.

EDUCATION

CentraleSupélec - Université Paris-Saclay

Engineering Degree

Paris-Saclay, France

Sep 2024 – Jun 2027

- Key courses: Measure Theory & Probability (Hilbert spaces, L^p spaces, conditional expectation, random walks), Partial Differential Equations (distributions, Sobolev spaces, variational formulation), Algebra & Cryptology (lattice reduction, LLL algorithm, elliptic curves, learning with errors), Information Theory (entropy, mutual information, f-divergences), Statistics & Machine Learning, Optimization (convexity, duality, Lagrange multipliers)

Lycée Hoche

CPGE: MPSI – MP

Versailles, France

Sep 2021 – Jun 2024

- Intensive two-year undergraduate program in mathematics and physics, preparing for nationwide competitive entrance exams to top French engineering schools (equivalent to a Bachelor's level in Mathematics).

PROFESSIONAL EXPERIENCE

Lycée Hoche - Éducation Nationale

Oral Examiner

Versailles, France

Sep 2024 – Jun 2026

- Assessed 6 second-year students weekly in 2-hour oral examinations ("khôlles") in mathematics, preparing tailored exercises for each student's level (see examples) and reporting individual progress to the professor.

Sumécatronic

Milling Machine Operator

Magny-Les-Hameaux, France

Jun 2025 – Jul 2025

- Autonomous milling machine operator, producing and quality-controlling series of precision-engineered parts for the aeronautics industry. Gained hands-on experience with quality assurance processes and workshop logistics.

PROJECTS

Course project: Robot Fault Diagnosis through Transfer Learning and Digital Twins — CentraleSupélec LGI | Gif-sur-Yvette, France

Contributor

Jan 2025 – Jun 2025

- Since failure data is scarce for mechanical arms (built not to fail), explored the feasibility of classifying real faults using a classifier trained on simulated data — a typical domain adaptation problem.
- Achieved 87% AUC on real robot fault detection using a DANN (Domain-Adversarial Neural Network) built on a 1D-CNN architecture.

Course project: Multi-view Sport Video Synchronisation — GoPro |

Gif-sur-Yvette, France

Contributor

Jan 2026 – Jun 2026

- Exploring methods for automatic multi-view sport video synchronisation, comparing three approaches: human pose detection, audio-based alignment using transformers, and visual similarity matching.

Course project: Variance Reduction for Monte Carlo Simulation in Financial Mathematics |

Gif-sur-Yvette, France

Contributor

Mar 2026

- Explored low-discrepancy sequences as alternatives to standard Monte Carlo methods and built a benchmark comparing all approaches on a grid of volatility, dimension, and moneyness using $|\text{err}| \sqrt{\text{time}}$ as a metric.
- Found that Sobol sequences were the most efficient method in 46% of configurations; trained a random forest achieving 81% CV accuracy to predict the best method for a given configuration.

LEADERSHIP & INVOLVEMENT

Handi-tutorat (in partnership with Sopra Steria)

Tutoring for students with disabilities

Gif-sur-Yvette, France

Jan 2025 – Present

- Volunteered to support students with disabilities facing academic difficulties, helping them develop personalized learning methods and integrate into campus life.

Private Lessons

Mathematics Tutor

Paris Area, France

Sep 2024 – Jun 2025

- Provided academic support for high schoolers in mathematics, focusing on building their mathematical intuition and fostering independence.

SKILLS

Programming: Python (NumPy, Pandas, scikit-learn, PyTorch), OCaml, SQL, MATLAB, R;

Tools: Git, Bash, Slurm, HPC;

Languages: French (native), English (advanced), German (intermediate).

INTERESTS

Golf: Regular player;

Harp: 10 years, 7 years in conservatoire;

Associations: Cooking association of CentraleSupélec (head of testing);